MIL-S-60661A (AR)
1 JUNE 1978

SUPERSEDING
MIL-S-60661 (MU)
10 FEBRUARY 1971

MILITARY SPECIFICATION

SIGHT, LEAD COMPUTING, M61, 11747062

This specification is approved for use by the U. S. Army Armament Research and Development command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a lead computing sight used on Gun, Anti-Aircraft **Artillery, Self-Propelled 20 MM, M163 and Towed 20 MM, M167.**

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-F-13926	**Fire control Material; General Specification Governing the Manufacture and Inspection of Parts, Equipment and Tools for Army Material; Packaging and Packing of**
MIL-P-14232	
MIL-I-45607	**Inspection Equipment, Acquisition, Maintenance and Disposition of**

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC **1285**

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U. S. Army, ARRADCOM

D11747062	Sight, Lead Computing
10545659	Shaft, Mounting
8436849	Mount, Shock
10545536	Inverter
10545270	Module, Magnet Current Supply
10545260	Module, Torque Current Supply

PACKAGING DATA SHEET

D11747062	Packaging of Lead Computing Sight

(Copies of specifications, standards, drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3, REQUIREMENTS

3.1 Fabrication. - The lead computing sight shall be manufactured in accordance with Drawing 11747062 and drawings pertaining thereto,

3.2 General Specifications. - Where applicable, the lead computing sight shall meet the requirements of MIL-F-13926 as follows:

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances

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c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.3 Environmental Service Conditions

3.3.1 Vibration. The sight shall be mounted in normal service position, on recoil absorbing mounts, utilizing mount kit parts 8436849 and shafts 10545659. The sight load shall meet the requirements of 3.5 within the degradation limits specified in 3.6 after being vibrated as specified below along three mutually orthogonal axes for at least three (3) hours per axis.

Amplitude	Frequency (Hz)
.076 in. Double Amplitude	5-23
2g	23-40
4g	40-120
2.5g	120-200
4g	200-500

The vibration frequency shall be cycled logarithmically with a 15-minute period. Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. With the sight mounted in a shock fixture utilizing mount kit part number 8436849 and shafts part number 10545659. The sight shall meet the requirements of 3.5 within the degradation limits specified in 3.6 after being subjected to a total of eighteen 10G, n-millisecond shocks, three shocks applied in opposite directions along each of the three mutually perpendicular axes of the sight. The point for application and measurement of shock shall be the test fixture .In addition the sight shall meet the requirements of 3.5 within the degradation limits specified in 3.6 after being subjected to shocks as specified above except the sight shall be mechanically caged and the shock magnitude shall be 15G's.

3.3.3 Temperature.

3.3.3.1 Operating. With the neutral density filter removed, the sight shall meet the requirements of 3.5 within the degradation limits specified in 3.6 when stabilized thermally at temperature extremes between -40°F and +155°F per the procedure of Paragraph 4.6.4.

3.3.3.2 Non operating. (storage) With the neutral density filter removed, the

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sight shall not suffer permanent damage when subjected to not less than 4 hours at each temperature extreme of -65°F and +155°F, and shall meet the requirements of 3.5 when returned to room ambient.

3.3.4 Electrically caged 18-hour capability. The sight shall not suffer damage when caged electrically for 18 hours, and shall meet the requirements of 3.5 at the completion of the 18 hours.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the sight.

3.5 Performance.

3.5.1 Definitions. The following signs, polarities, and definitions are followed throughout this specification:

B	=	rate "+" = CCW rotation of sight case.
T	=	Traverse lead angle "+" = reticle right
E	=	Elevation lead angle "+" = reticle down
GT	=	Gyro traverse angle "+" = reticle right
GE	=	Gyro elevation angle "+" = reticle down.
eT	=	Traverse lead angle error
eE	=	Elevation lead angle error
eR	=	Resultant lead angle error
	=	$[(\lambda_{eE})^2 + (\lambda_{eT})^2]^{1/2}$
Vt	=	input to Torque current module 10545260

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Vm = input to Magnet current module 10545270

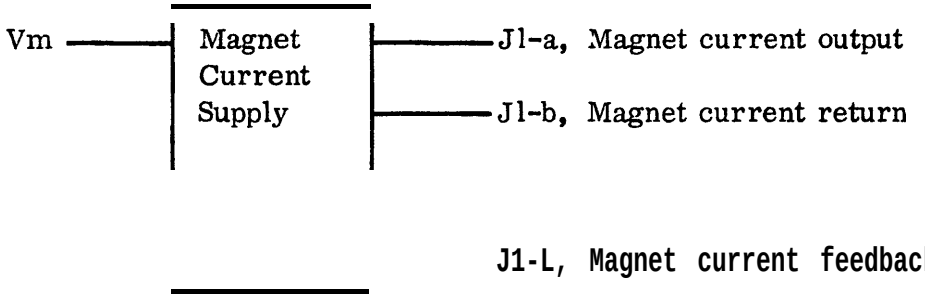
3. 5.2 Electrical Input Requirements. With following power requirements are necessary in order to perform the tests contained herein.

3.5.2.1 Gyro motor excitation applied to
J1-f,g,h .
l15 ± 5~VRiMS, three phase, delta
400 ± l% Hz
Phase rotation J1-f,g,h
Starting current, 3 amps average per phase, maximum
Running current, 0.75 amps average per phase, maximum

3.5.2.2 Firing indicator lamp, J1-e
22 to 30 VDC
Current, 50 MA, maximum
Lamp return Jl, j

3.5.2.3 Reticle lamp, major J1-Y, minor J1-d
13 VDC
Current, 1.6 amps, maximum
Lamp return J1-M

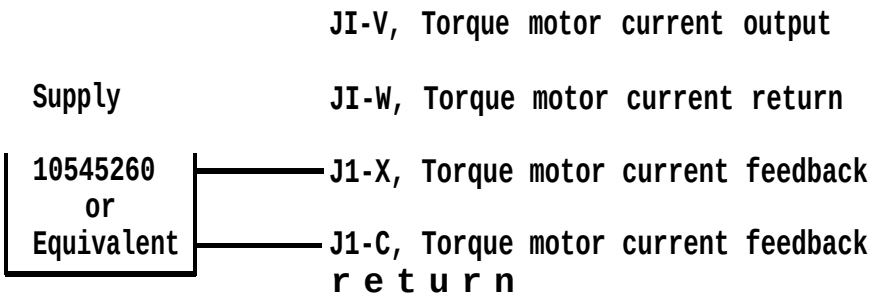
3.5.2.4 Magnet Current. -
0 to 700 ma.
Magnet current supply, part number 10545270 or electrically equivalent circuitry, may be used to supply magnet current per the following diagram.



3.5.2.5 Torque Current. -
0 to ± 555 ma.
Torque current supply, part number 10545260 or electrically equivalent circuitry, may be used to supply torque current per

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the following diagram.



3.5.3 Output Voltages.

3.5.3.1 Elevation Gimbal Angle (X_{GE}) Synchro Transmitter*

Range: ± 31 degrees
Voltage: 90 volts line to line
Frequency: 400 Hz
Phasing:
= O D-E and F-E shall be in phase with f-g.
O D-F shall be in phase with f-g for the down reticle condition.

3.5.3.2 Traverse Gimbal Angle ($\sim_{GT}$) Synchro Transmitter.

Range: ± 31 degrees
Voltage: 90 volts line to line
Frequency: 400 Hz
Phasing:
A = O P-R and S-R shall be in phase with f-g.
k~~ # O p-s shall be in phase with f-g for the right reticle condition.

3.5.4 Sight Reticle.

3.5.4.1 Image. - The sight reticle image shall be red and shall be composed of two illuminated, concentric circles 60 ± 3 mr and 15 ± .5 mr in diameter respectively. The 15 mr circle shall contain a 3 ±. 5 mr gap at the six o'clock(± 5°) position.

3.5.4.2 Parallax. The apparent displacement of the image of the inner reticle circle shall not exceed 1.0 mr when viewed from points 1.0 inch apart on the major

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traverse and elevation axes.

3.5.4.3 Lamp. The reticle shall be illuminated by a dual-filament lamp. Current required per filament shall be between 1.0 and 1.6 Amps at 12.6 Vdc.

3.5.5 Ready to Fire Light. The ready to fire light shall require 50 ma maximum at 28 volts dc.

3.5.6 Gyro Motor. With no magnet current applied and the sight caged, the gyro shall come up to speed within 15 seconds. The maximum starting current shall be 3.0 amps per phase. The running current shall be less than 0.75 amps per phase. The current shall be monitored at pins f and h.

3.5.7 Mechanical Caging. The mechanical cage locks the gyro in a position corresponding to the traverse lead angle AT= 0 ± 2 mr (%T ÷O ± 0. 14°) and the elevation lead angle ~E=+7 ± 2 mr (~E= .50 ± .14°). An interlock switch shall be actuated by the caging mechanism in the cage position.

3.5.8 Electrical Caging. A magnet current of 700 ± 50 ma (Vm = 13.17 ± . Olv) electrically cages the sight in a position corresponding to a traverse lead angle of AT = 0 ± 1.2 mr ($\sim_{GT}$=O ± . 070) and an elevation lead angle $\sim_{E}$= O ± 1.2 mr (A~~ =GE÷0 ± 07°). In the electrically caged position the gyro axis shall be parallel both to the gyro mounting surfaces and to the plane of the mounting bolts within ± 2 mr. The center lead angle indicator on the traverse gimbal shall be aligned with the pointer.

30509 Gimbal Freedom. The traverse and elevation gimbal freedom shall be at least ± 31°.

3.5010 Lead Angle Accuracy. The sight shall be tested for each of the points in Table I. The tolerance shall be met for at least two-thirds of the points in Table I, and the tolerance shall not be exceeded by more than 1.0 mr at any point. For turning rates of 45°/second or greater, the correct lead angle reading is the initial lead angle reading within one revolution of the rate table.

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TABLE I. Lead angle accuracy,

SIGHT CASE ATTITUDE	RATE DEGREES/SEC. ± .01°/sec.	Vm VOLTS ±.01v	Vt VOLTS ±01v	NOMINAL LEAD ANGLE TRAVERSE Degrees	TRAVERSE mr	ELEVATION Degrees	ELEVATION mr	RADIAL ERROR mr
Step 1								
1	+0.57	6.04	0	0.57	10.0	0	0	3.1
2	+5.73	3.58	0	17.46	304.7	0	0	7.4
3	+11.46	6.04	0	11.54	201.4	0	0	5.9
4	-11.46	4.34	0	23.58	411.5"	0	0	11.0
5	+22.92	13.17	0	4.59	80.1	0.	0 '	4.1
6	+34.38	13.17	0	6.89	120.3	0	0	4.7
7	+45.84	13.17	0	9.21	160.7	0	0	5.3
8	+57.30	13.17	0	11.54	201.4	0	0	5.9"
9	-0.57	2.96	0	-2.58	-45.0	0	0	3.6
10	-2.87	3.33	0	-10.08	-179.5	0	0	5.5
11	-5.73	3.33	0	-20.49	-357.6	0	0	8.1
12	-11.46	4.34	0	-23.58	-411.5	0	0	11.0
13	-17.19	8.44	0	-8.63	-150.6	0	0	5.2
14	-28.65	13.17	0	-5.74	-100.2	0	0	4.4
15	-45.84	13,17	0	-9.21	-160. ?	0	0	5.3
16	+2.87	4.34	-0.09	0	0	5.74	100.2	4.4
17	+5.73	3.33	-2.24	0	0	20.49	357.6	8.1
18	+17.19	8.44	-1.17	0	0	8.63	150.6	5.2
19	+57.3	13.17	-6.99	0	0	"11.54	201.4	5.9
20	-2.87	4.34	+0.09	0	0	-5.74	-100.2	4.4
21	-5.73	3.33	+2.24	0	0	-20.49	-357.6	8.1 "
22	-17019	8.44	+1.17	0	0	-8.63	-150.6	5.2
23	-57.3	13.17	+6.99	0	0	-11.54	-201.4	5.9

3.5.11 Magnet Current. Using direct current (not magnet current module), the magnet current required to produce lead angles for given turning rate shall be:

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RATE Deg. /See. ±. 03 °/See.	LEAD ANGLE mr +1	LEAD ANGLE Degrees +. 06	Im Amps
+2. 86	100	+5. 73	.223 to .234
+5. 72	100	+5. 73	.313 to .329
+8. 47	300	+17,19	.223 to .234
+16. 93	300	+17. 19	.313 to .329
-2.86	100	-5.73	.223 to .234
-5.72	100	-5.73	.313 to .329
-8.47	300	-17019	.223 to .234
-16.93	300	-17.19	.313 to .329

3.5.12 Superelevation. With the sight stationary, the voltage imputs to the magnet current module and to the torque current module to produce superelevation angles shall be:

V_t volts +. Olv	V_m volts +. Olv	B Deg. /See.	Lead Angle Traverse mini rad.	Lead Angle Elevation mini rad.
-.20	2.96	0	0 ± 2	10+2
-.41	2.96	0	0 ± 2	20 ± 2
-,71	2.96	0	0+2	35 ± 2
-1.42	2.96	o "	0 ± 2	70 ± 2

3.5013 Gyro Pickoff. The lead angles required to produce the gyro pickoff angles shall be:

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Gyro Pickoff Angle Degrees ± . 03° Traverse	Lead Angle mini rad.	Lead Angle Degrees	Ref Only Vm volts	Ref Only B Degrees/Sec.
5°CCW	(+) 68.5- 71	3.90- 4.07	8.44	+7. 81
10°CCW	(+) 137 - 142	7.85- 8.14	8.44	+15. 63
25°ccw	(+) 343.5-355	19.68- 20.34	4.97	+13. 38
355°cw	(-) 68.5- 71	356.10-355,93	8.44	-7.81
350°cw	(-) 137 - 142	352.15-351.86	8.44	-15.63
335°cw	(-) 343.5- 355	340.32-339.66	4.97	-13.38
Elevation				
5°CCW	(+) 68. 5- 71.	3.90- 4.07	8.44	+7. 81
10°ccw	(+) 136.5 -141.5	7.82- 8.14	8.44	+15. 63
25°CCW	(+) 334 -345	19.14- 19.77	4.97	+13. 38
355°cw	(-) 68.5- 71.	356.10-355.93	8.44	-7.81
350°cw	(-) 136.5 -141.5	352.18-351.89	8.44	-15.63
335°cw	(-) 335 -346	340.81-340.18	4.97	-13.38

3.5.14 Lead Angle Indication. The pointer shall line up with the lead angle indicators on the traverse gimbal for the following traverse synchro angles.

Lead Angle Indicator .	Synchro Angle Degrees ± . 060)	Vm Volts (ref. only)	B Degrees/Sec. (ref. only)
+100 mr (inner dot)	7.16	8.44	-11.25
+200 mr (outer dot)	14.32	8.44	-22.76
-100 mr (inner dot)	352.84	8.44	+11. 25
-200 mr (outer dot)	345.94	8.44	+22.76

305015 Pressure. The sight shall be pressurized to 5-1 pounds per square inch guage of dry nitrogen. The leakage shall be less than 0.2 PSIG per hour. when pressurized at 15 PSIG.

3.6 Extended Limits. Where extended limits are specified, the lead angle error of Table I shall be modified as shown in Table I and in accordance with the following:

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The allowable lead error for 16 of the lead angles tested at selected points in Table I shall not exceed $4 \pm^{A}_{R}$ mr where R is the nominal lead angle magnitude in degrees as specified in Table I. In addition no single lead angle tested shall be in excess of $5 \pm \angle @ mr_c$

4, QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) submission. - The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of one sight manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provision.

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TABLE II. Lead angle accuracy (extended limits).

SIGHT CASE ATTITUDE	RATE DEGREES/SEC. ±. ol°/sec.	Vm ±	Vt .olv	NOMINAL LEAD ANGLE (DEG) (Synchro) TRAV	NOMINAL LEAD ANGLE (DEG) (Synchro) ELEV	GIMBAL RADIAL ERROR (DEGREES)	OPTICAL ERROR (MR)
	± 0. 57	6.04	0	0.72	0	•31	4.28
	+5. 73	3.58	0	21.83	0	.91	12.73
	+11. 46	6.04	0	14.43	0	.70	9.77
	+11. 46	4.34	0	29.48	0	-1.13	15.79
	+22.. 92	13.17	0	5.74	0	.45	6.30
	+34. 38	13.17	o	8.62	0	•53	7.45
	+45, 84	13.17	0	11.51	0	.62	8.61
	+57. 30	13.17	0	14.43	0	.70	9.77
	-0.57	2096	0	-3.23	0	.38	5.29
	-2.87	3.33	0	-12.60	0	•65	9.04
	-5.73	3.33	0	-25.61	0	1.02	14.25
	-11.46	4.34	0	-29.48	0	1.13	15.79
	-17.19	8.44	0	-10.79	0	.60	8.32
	-28.65	13.17	0	-7.18	0	•49	6.87
	-45.84	13.17	0	-11.51	0	.62	8.61
	+2. 87	4.34	-0.09	0	7.20	•49	6.87
	+5. 73	3.33	-2.24	0	26.04	1.04	14.25
	+17 •19	8.44	-1.17	0	10.82	.60	8.32
	+57 .3	13.17	-6.99	0	14.51	.70	9.77
	-2.87	4.34	+0. 09	|	-7.20	.49	6.87.
	-5073	3.33	2.24	0	-25.88	1004	14.25
	-17.19	8.44	+1. 17	0	-10.82	.60	8.32
	-57.3	13017	+6. 99	0	-14.52	.70	9.77

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4.3.1 Submission of ~roduct. Unless otherwise s~ec ified bv the contracting officer, inspection lot size, lot }ormation and presentation o; lots sha~l be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and Tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provision (SQAP) listed in the Technical Data Package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 and 4.4 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.3. 2.3 and 4.4 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests. The requirements and tests in Table III shall be inspected on a 100 percent basis. All examination and tests shall be conducted at the standard ambient temperature (60° F to 90° F).

TABLE III. - Classification of defects
(100 percent inspection)

CHARACTERISTIC	Requirement	TEST PROCEDURE
CRITICAL: NONE		
MAJOR:		
101. Electrical	3.4	4.6.2
102. Performance	3.5	4.6.3
MINOR: NONE		

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance

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of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special Sampling. - One sight shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table IV. One sight shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table V.

TABLE IV. - Special sampling "

CONTROL TEST		REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawings-Visual
302.	General Specifications	3.2	MIL-F-13926 - Visual

TABLE V. - Special sampling, environmental

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
303.	Vibration	3.3.1	4.6.1.1
304.	Shock	3.3.2	4.6.1.2
305.	Temperature	3.3.3	4.6.1.3

4.4.1 Failure of Sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3. 3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL- I-45607

4.5.1 Government - furnished inspection equipment. - Where the contract provides for Government - furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607

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4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the test product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4*5. 2.3 Test Equipment. - In conjunction with 4.5.2.2, the test equipment listed in Table VI shall be utilized in performing the applicable test of 4.6.

4.6 Test methods and procedures.

4.6.1 Environmental. - Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810, except 'as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

TABLE VI - Test equipment list

ITEM NUMBER		DESCRIPTION
1.	Test Console	Cabinet to house metering equipment and control panel (see figure 1)
2.	Power Supply	DC, 28 volts, unregulated 27 amps
3.	Power Supply	DC, +15 volts, regulated 300 ma

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TABLE VI - Test equipment (cont'd)

ITEM NUMBER		DESCRIPTION
4.	Power Inventor	3 phase, 400 cycle, P/N 10545270
5.	"Magnet Current Module	P/N 10545270
6.	Torque Current Module	P/N 10545260
7.	Ammeter	AC, 5 amps 2%
8.	Voltmeter	DC, digital, 30 volts +. 05%
9.	Angle Position Indicator	Accuracy + .03 degrees
100	Reticle Lamp Supply	± 12 volts, adjustable
11.	Magnet Current Supply	± 12 volts, adjustable
12.	TV Monitor	15 MMZ bandwidth 2% geometric linearity
13.	Rate Table	+60°/sec~ accuracy of 1/2% indicated speed. 30 slip rings input, 5 ft. /lbs. peak torque
14.	Sight Holding Fixture	Calibrated rotatable carriage with two axis capability. + MR accuracy and synchro angle readout. (see Figure 2)
15.	Collimator	F. L. - 300-400mm with calibrated grid pattern of lMR
16.	TV Camera/With Lens	500 Line center resolution, 175mm lens
17.	Boresight Calibration Fixture	+1/2 MR accuracy with 5x min. reticle alignment telescope
18.	Control Panel	Contains switches, gauges and control switches
19.	Switch	Single Pole, (SP) Triple throw or rotary

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TABLE VI. - Test equipment (cont'd)

ITEM NUMBER		DESCRIPTION
20,	Switch	SP, Single throw (ST)
21.	Switch	SP, Double throw
22.	Switch	SP, ST, Momentary Contact, Push-Button
23.	Resistor	1, Ohm, 10 watt l%
24.	Switch	Single Pole, 6 position Rotary
25.	Switch	Single Pole, Double throw, Center off
26.	Variable Resistor	750 Ohms, 2 watt 10%
27.	Variable Resistor	50 Ohms, 2 watt, 10%
28.	Variable Resistor	1K Ohms, 2 watt, 10%
29.	Variable Resistor	200 Ohm, 2 watt, 10%

4.6.1.1 Vibration. The test shall be performed as specified in method 514.2 Procedure VIII of MIL-STD-810 as modified by 3.3.1 herein. After vibration, the sight shall be tested as specified in 4.6.3 to determine compliance with 3.3.1.

4.6.1.2 Shock. This test shall be performed as specified in Procedure I, method . 516.2 of MIL-STD-810 as modified by 3. 3.2 herein. After shock impulses, the sight shall be tested as specified in 4. 6.3 to determine compliance with 3.3.2.

4.6.1.3 Temperature

4.6.1.3.1 High (Non-Operating and Operating). This test shall be performed as specified in Procedure I, Method 501 of MIL-STD-810, as modified by 3. 3.3 herein. The temperature of Step 2 shall be +155°F. The temperature of Step 3 shall be maintained for not less than 4 hours. The temperature of Step 4 shall be +155°F. In Step 5, the temperature shall be maintained for 4 hours and the sight shall be tested in accordance with 4.6.4 herein. In Step 7 the sight shall be tested in accordance with 4.6.3 to determine compliance with 3.3.3.

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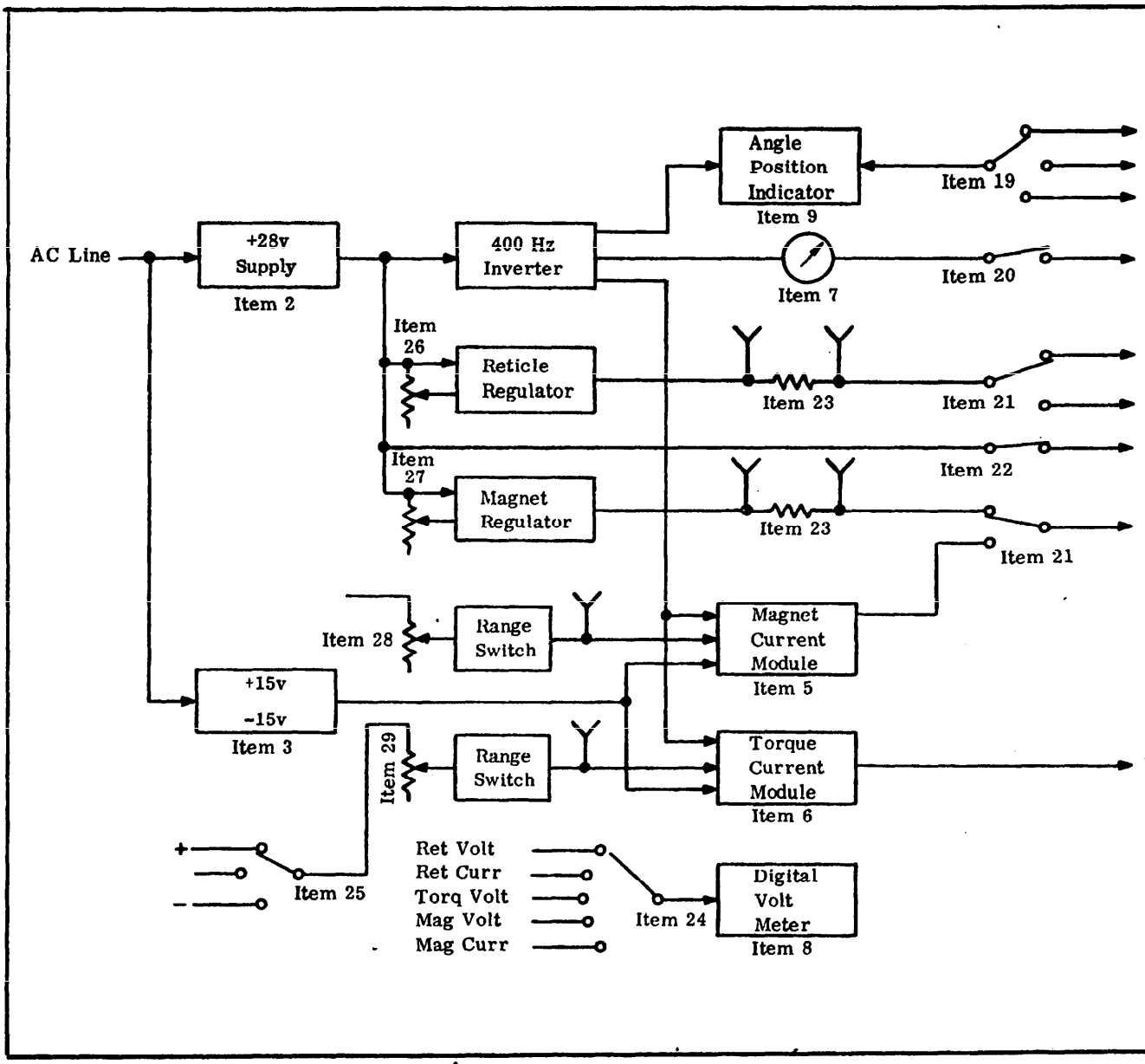


Figure 1. – Sight Test Console

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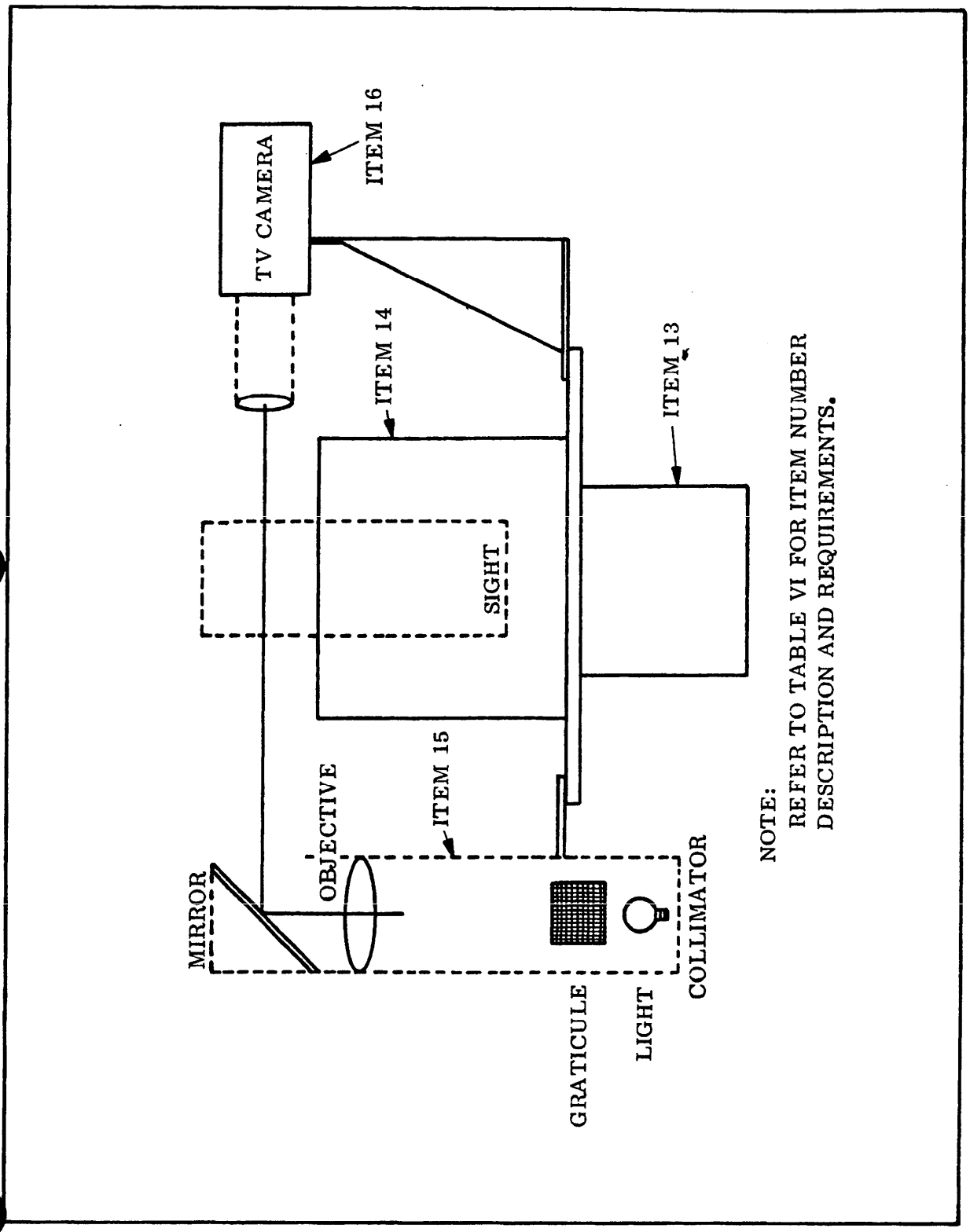


Figure 2. - System Test Set-Up

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4.6.1.3.2 Low (Non-Operating and Operating). This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810 as modified by 3.3.3 herein. The temperature of Step 2 shall be -65°F maintained for not less than 4 hours. The temperature of Step 4 shall be -40°F. In Step 5, the sight shall be tested in accordance with 4.6.4 herein. In Step 7, the sight shall be tested in accordance with 4.6.3 to determine compliance with 3.3.3.

4.6.1.4 18 hour cage. The sight shall be operated for not less than 18 hours with 700 ± 50 ma. current through the magnet coil. After the cage test, the sight shall be tested as specified in 4.6.3 to determine compliance with 3.3.4.

4.6.2 Electrical. Continuity shall be inspected by Standard Measure Equipment (SME), and visual examination for compliance with 3.4.

4.6.3 Performance. The sight shall be tested mounted in the Sight Holding Fixture. This fixture, and its associated TV camera and collimator are mounted on the rate table. The Sight Test Console provides the necessary inputs and monitoring equipment for the sight.

4.6.3.1 Sight Reticle. - Apply power as per 3.5.2.3 to the major filament of the reticle lamp. Adjust the intensity of the lamp, using the dimmer control on the Sight Test Console, to get a sharp picture of the reticle on the television monitor. Adjust the intensity of the illuminated collimator for a sharp grid pattern on the television monitor. Set the movable carriage to the straight ahead (bore sight) position.

4.6.3.101 Measure the traverse and elevation diameters, the width of the gap, and the location of the gap for the inner circle. These dimensions shall be as specified in 305.401.

406.3.1.2 Measure the diameter of the outer circle by adjusting the movable carriage so the left sides of both the outer and inner circles are visable. Measure the distance between the two circles. Move the carriage to the other side and measure between the inner and outer circles. The diameter of the outer circle is the sum of these two distances and the diameter of the inner circle. The diameter of the outer circle shall be as specified in 3.5.4.1.

. 4.6.3. 1.3 Rotate the sight holding fixture so connector J1 is down. Repeat 4.6.3..1.2.

4.6.3.1.4 Move television camera up 1 inch, and then return to rest position. The motion of the reticle relative to the grid shall be as in 3.5.4.2

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4.6.3.1,5 Return sight to upright position and repeat 4.6.3.1.4.

406.3.1.6 Adjust reticle voltage to 12.6v. dc. Measure the major filament current. Switch to the minor filament, and readjust reticle voltage to 12.6 vdc. Measure the minor filament current. Current for both filaments shall be as specified in 3.5.4.3. Readjust reticle voltage for sharp reticle display on television monitor.

4.60302 Ready to Fire Lamp. - Apply power to the Ready to Fire Lamp as specified in 3.5.2.2. The lamp shall light and the current shall be as specified in 3.5.5.

4.6.3.3 Gyro Motor. - Apply power to the gyro motor as specified in 3.5.2.10 The starting current into pins f and h, the time it takes the gyro to come up to speed as indicated by a drop in current, and the running current into pins f and h shall be as specified in 3.5.6.

4.6.3.4 Mechanical Cage. - Mechanically cage the sight by turning the cage knob fully clockwise With the movable carriage set to straight ahead (bore sight position), the traverse and elevation displacement of the center of the reticle with respect to the boresight center of the grid pattern shall be as specified in 3.5.70 Measure the traverse and elevation synchro angles. These angles shall be as specified in 305.7.

4.6.3.5 Electrical Cage. - Mechanically uncage the sight by turning the cage knob fully counter-clockwise. Electrically cage the sight by applying Vm=13. 17 volts to the magnet current module. The displacement of the center of the reticle with respect to the boresight center of the grid pattern shall be as specified in 3. 5• 8. Measure the traverse and elevation synchro angles. These angles shall be as specified in 3.50 8• Check the alignment of the pointer with the center lead angle indicator on the traverse gimbal-check the alignment of the gyro axis with the gyro mounting surfaces and the plane of the mounting bolts as specified in 305.8.

4.6.306 Gimbal Freedom.

4.6.3.601 Set movable carriage to boresight position.

406.306.2 Electrically uncage gyro by setting Vm=O.

4.60306.3 Manually rotate movable carriage CCW until the traverse synchro angle stops increasing~ This angle shall be as specified in 305090

4.6.306.4 Electrically cage gyro by setting Vm=13. 17 volts. Return movable carriage to boresight position~

4.6.30605 Uncage gyro by setting Vm=O volts. Manually rotate movable carriage

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CW until traverse synchro angle stops decreasing. This angle shall be as specified in 3.5.9.

4.6.3.6.6 Electrically cage gyro by setting VM=13. 17 volts. Rotate sight around its optical axis so input connector J1 is down.

4.6.3.6.7 Set movable carriage to straight ahead position.

4.6.3.6.8 Electrically uncage sight by setting Vm=O. Manually rotate movable " carriage CCW until the Elevation Synchro Angle stops increasing. This angle shall be as specified in 3.5.8.

4.6.306.9 Electrically cage gyro by setting Vm=13. 17 volts. Return movable carriage to straight ahead positon.

4.6.3.6.10 Uncage gyro by setting Vm=O. Manually rotate movable carriage CW until the elevation synchro angle stops decreasing. This angle shall be as specified in 3.5.8.

4.6.3.6.11 Electrically cage sight by setting Vm=13. 17 volts. Return sight to. upright position.

4.6.3.7 Lead Angle

4.6.3.7.1 Traverse Lead Angle. - Set the movable carriage to an angle equal and opposite to the desired lead angle. Set in the values of rate and Vm from Table I, steps 1 to 15. The displacement of the center of the reticle from the boresight center of the graph is the error. The resultant of the elevation and traverse errors shall be as specified in 3.5.10.

4.6.3.7.2 Elevation Lead Angle. - Rotate the sight about its optical axis 3° connector J1 is down. Set the movable carriage to an angle equal and opposite to the desired lead angle. Set in the values of rate, Vm, and Vt from Table I, steps 16 to 23. The displacement of the center of the reticle from the boresight center of the graph is the error. The resultant of the elevation and traverse errors shall be as specified in 3.5.10. Return the sight to its upright position.

4.6.3.8 Magnet Current. - Electrically disconnect the magnet current module from the sight. Connect a one ohm resistor between ill-a and ground. Apply a dc current into J1-b. the voltage drop across the one ohm resistor is equal to the magnet current in Amps. Apply the rates and adjust the magnet current to get the lead angles specified in 3.5.11. The current shall fall within the limits of 3.5.11.

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4.6.3.9 Superelevation. - Reconnect the magnet current module. With the sight being stationary, apply the magnet module input voltage, Vm, and the torque module voltages, Vt, as specified in 3.5.12. The elevation and traverse lead angles shall be as specified in 3.5.12.

4.6.3.10 Gyro Pickoffs

4.6.3 .10.1 Traverse Pickoffs

4.6.3 .10.2 Apply the rate and Vm as suggested in 3.5.13. Adjust the rate to get the required traverse gyro angle.

4. 6.3.10.3 Adjust the movable carriage to center the reticle horizontally on the boresight center of the graph. The angle of the movable carriage with respect to the boresight position of the carriage is equal and opposite to the lead angle specified in 3.5.13.

4.6.3 .10.4 Elevation Pickoff

4.,.3.,0.5 Rotate the sight about its optical axis so connector J1 is down. Apply the rate and Vm as suggested in 3.5.13. Adjust the rate to get the required elevation gyro angle.

4.6.3 .10.6 Adjust the movable carriage to center the reticle horizontally on the boresight center of the graph. The angle of the movable carriage with respect to the boresight position of the carriage is equal and opposite to the lead angle specified in 3.5.13. Return sight to upright position.

4.6.3.11 Lead Angle Indication. - Apply Vm and rate as suggested in 3.5.14. Adjust rate for specified Traverse Synchro angle. Pointer shall be aligned with dot specified in 3.5.14.

4.6.3.12 Pressure. - Pressurize sight to 152,1 PSIG. After 1 hour the pressure shall not have decreased by more than O. 20 PSIG. Reduce pressure in sight to 5^{+0}_{-1} PSIG.

4.6.4 Environmental Operating Procedure. - The following procedure shall be used for checking operation of the sight in environmental chambers. Previous to environmental testing of the sight, the elevation and traverse pickoff synchros shall have been calibrated over the operating range specified in 3.3.3.1. The lead angles shall be determined from the traverse and elevation pickoff synchros. The environmental chamber may be opened for 10-15 minutes to uncage the sight or other mechanical adjustments.

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4.6.4.1 Each of the points in Table II shall be tested. The limits in Table II reflect the extended limits of 3.6. The resultant error of at least 16 of the points shall fall within the error limit of Table II, and the rest shall not exceed the error limit of Table II by more than 0.07 degrees.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 117. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended Use. - The sight is used in the Anti-Aircraft Artillery M163 and M167 to track line-of-sight moving targets and to provide lead angles for gun direction and position outputs for radar pointing.

6.2 ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation, packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

6.3 Previous issue. - The basic issue of this specification was released only for a specific application and is not available from the DoD Single Stock point.

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